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Instructions Follow instructions *carefully*; failure to do so will result in points being deducted. You may write or type out your answers. Be sure to show sufficient work to justify your answer(s).

1. 6 points Determine if each of the following sentences is a proposition. Justify your answer.
- (a) The number 17 is a prime number.
 - (b) $5 + 8 = 12$
 - (c) Close the door.

Solution: Enter your solution here:

A: It is a proposition, because it is a declarative statement, and has a definite truth value.

B: It is a proposition, because it is a declarative statement, and has a definite truth value.

C: It is not a proposition, because it is a command and not a declarative statement, and has no definite truth value.

2. 12 points Let the following propositions be defined as:

- d : It's Valentine's Day.
- e : You go on a Valentine's Day excursion.
- a : You eat chocolate and ice cream.

Express each of these compound propositions as an English sentence.

- (a) $d \rightarrow a$
- (b) $\neg e \rightarrow a$
- (c) $(d \rightarrow e) \wedge (d \rightarrow a)$
- (d) $(a) \leftrightarrow (d \wedge e)$

Solution: Enter your solution here:

A: If it's Valentine's Day then you eat chocolate and ice cream.

B: If you do not go on a Valentine's Day excursion then you eat chocolate and ice cream.

C: If it's Valentine's Day then you go on a Valentine's Day excursion and if it's Valentine's Day then you eat chocolate and ice cream.

D: You eat chocolate and ice cream if and only if it's Valentine's Day and you go on a Valentine's Day excursion.

3. 6 points Let the following propositions be defined as:

- w : You wore your winter coat.
- v : You visited an ice sculpture garden.

- d : You drank hot cocoa.

Write these propositions using w , v , and d and logical connectives (including negations).

- If you visited an ice sculpture garden, then you wore your winter coat and drank hot cocoa.
- If you didn't wear your winter coat, then you didn't visit the ice sculpture garden.

Solution: Enter your solution here:

A: $v \rightarrow (w \wedge d)$

B: $\neg w \rightarrow \neg v$

4. 16 points Show that the following conditional statement is a tautology by using a truth table.

- $[a \wedge (a \rightarrow b)] \rightarrow b$
- $(a \wedge b) \rightarrow (b \rightarrow a)$
- $[(a \vee b) \wedge (a \rightarrow c) \wedge (b \rightarrow c)] \rightarrow c$
- $[\neg a \wedge (a \vee b)] \rightarrow b$

Solution: Enter your solution here:

A:

a	b	$a \rightarrow b$	$a \wedge (a \rightarrow b)$	$[a \wedge (a \rightarrow b)] \rightarrow b$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

B:

a	b	$a \wedge b$	$b \rightarrow a$	$(a \wedge b) \rightarrow (b \rightarrow a)$
T	T	T	T	T
T	F	F	T	T
F	T	F	F	T
F	F	F	T	T

C:

a	b	c	$a \vee b$	$a \rightarrow c$	$b \rightarrow c$	$(a \vee b) \wedge (a \rightarrow c) \wedge (b \rightarrow c)$	$\rightarrow c$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

D:

a	b	$\neg a$	$a \vee b$	$\neg a \wedge (a \vee b)$	$\rightarrow b$
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	F	F	T

5. 16 points Prove by logical equivalences (without using a truth table) the following is a tautology. To get full credit, you must state which logical equivalences you used in each step.

- (a) $[a \wedge (a \rightarrow b)] \rightarrow b$
 (b) $(a \wedge b) \rightarrow (b \rightarrow a)$
 (c) $[(a \vee b) \wedge (a \rightarrow c) \wedge (b \rightarrow c)] \rightarrow c$
 (d) $[\neg a \wedge (a \vee b)] \rightarrow b$

Solution: Enter your solution here:

A:

$[a \wedge (a \rightarrow b)] \rightarrow b$	Implication
$[a \wedge (\neg a \vee b)] \rightarrow b$	Implication
$\neg[a \wedge (\neg a \vee b)] \vee b$	De Morgan
$[\neg a \vee \neg(\neg a \vee b)] \vee b$	De Morgan
$[\neg a \vee (\neg \neg a \wedge \neg b)] \vee b$	Double Negation
$[\neg a \vee (a \wedge \neg b)] \vee b$	Associative
$\neg a \vee (a \wedge \neg b) \vee b$	Associative
$\neg a \vee [(a \wedge \neg b) \vee b]$	Distributive
$\neg a \vee [(a \vee b) \wedge (\neg b \vee b)]$	Negation
$\neg a \vee [(a \vee b) \wedge T]$	Identity
$\neg a \vee (a \vee b)$	Associative
$(\neg a \vee a) \vee b$	Negation
$T \vee b$	Domination
T	Ends as true, proves tautology

B:

$(a \wedge b) \rightarrow (b \rightarrow a)$	Implication
$\neg(a \wedge b) \vee (b \rightarrow a)$	Implication
$\neg(a \wedge b) \vee (\neg b \vee a)$	De Morgan
$(\neg a \vee \neg b) \vee (\neg b \vee a)$	Associative
$\neg a \vee \neg b \vee \neg b \vee a$	Idempotent
$\neg a \vee \neg b \vee a$	Associative
$\neg a \vee a \vee \neg b$	Negation
$T \vee \neg b$	Identity
T	Ends as true, proves tautology

C:

$[(a \vee b) \wedge (a \rightarrow c) \wedge (b \rightarrow c)] \rightarrow c$	Implication
$[(a \vee b) \wedge (\neg a \vee c) \wedge (\neg b \vee c)] \rightarrow c$	Implication
$\neg[(a \vee b) \wedge (\neg a \vee c) \wedge (\neg b \vee c)] \vee c$	De Morgan
$[\neg(a \vee b) \vee \neg(\neg a \vee c) \vee \neg(\neg b \vee c)] \vee c$	De Morgan
$[\neg(a \vee b) \vee (\neg\neg a \wedge \neg c) \vee (\neg\neg b \wedge \neg c)] \vee c$	Double Negation
$[\neg(a \vee b) \vee (a \wedge \neg c) \vee (b \wedge \neg c)] \vee c$	Associative
$\neg(a \vee b) \vee (a \wedge \neg c) \vee (b \wedge \neg c) \vee c$	Distributive
$\neg(a \vee b) \vee [(a \vee b) \wedge \neg c] \vee c$	Associative
$[\neg(a \vee b) \vee [(a \vee b) \wedge \neg c]] \vee c$	Distributive
$[\neg(a \vee b) \vee (a \vee b)] \wedge [\neg(a \vee b) \vee \neg c] \vee c$	Negation
$T \wedge [\neg(a \vee b) \vee \neg c] \vee c$	Identity
$[\neg(a \vee b) \vee \neg c] \vee c$	Associative
$\neg(a \vee b) \vee (\neg c \vee c)$	Negation
$\neg(a \vee b) \vee T$	Domination
T	Ends as true, proves tautology

D:

$(\neg a \wedge (a \vee b)) \rightarrow b$	Implication
$\neg(\neg a \wedge (a \vee b)) \vee b$	De Morgan
$(\neg\neg a \vee \neg(a \vee b)) \vee b$	Double Negation
$(a \vee \neg(a \vee b)) \vee b$	De Morgan
$(a \vee (\neg a \wedge \neg b)) \vee b$	Distributive
$((a \vee \neg a) \wedge (a \vee \neg b)) \vee b$	Distributive
$(b \vee (a \vee \neg a)) \wedge (b \vee (a \vee \neg b))$	Negation
$(b \vee T) \wedge (b \vee (a \vee \neg b))$	Domination
$T \wedge (b \vee (a \vee \neg b))$	Associative
$T \wedge (a \vee (b \vee \neg b))$	Negation
$T \wedge (a \vee T)$	Domination
$T \wedge T$	Idempotent
T	Ends as true, proves tautology

6. 8 points Show that the following are logically equivalent without using a truth table. To get full credit, you must state which logical equivalences you used in each step.

(a) $\neg a \rightarrow (b \rightarrow c) \equiv b \rightarrow (a \vee c)$

(b) $\neg(\neg a \wedge b) \wedge (a \vee b) \equiv a$

Solution: Enter your solution here:

A:

$\neg a \rightarrow (b \rightarrow c) \equiv b \rightarrow (a \vee c)$	Implication
$\neg\neg a \vee (b \rightarrow c) \equiv \neg b \vee (a \vee c)$	Double Negation
$a \vee (b \rightarrow c) \equiv \neg b \vee (a \vee c)$	Implication
$a \vee (\neg b \vee c) \equiv \neg b \vee (a \vee c)$	Associative
$\neg b \vee (a \vee c) \equiv \neg b \vee (a \vee c)$	

B:

$$\neg(\neg a \wedge b) \wedge (a \vee b) \equiv a$$

De Morgan

$$(\neg\neg a \vee \neg b) \wedge (a \vee b) \equiv a$$

Double Negation

$$(a \vee \neg b) \wedge (a \vee b) \equiv a$$

Distributive

$$a \vee (b \wedge \neg b) \equiv a$$

Negation

$$a \vee F \equiv a$$

Identity

$$a \equiv a$$